The dual trinity theorem

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Abstract

In this paper, I describe the dual trinity theorem and prove it.
The paper ends with "The End"

The dual trinity theorem

Theorem 1. *For reals x, y and z, with $s = x + y + z$,*

$$2(x+y+z)\,e^{x+y+z} = \sum_{n=1}^{\infty}\sum_{r=0}^{\infty}\sum_{k=0}^{n} \frac{\binom{n+r-1}{r}\binom{n}{k}}{(n+r-1)!}\Big(x^r y^k z^{n-k} + x^k y^{n-k} z^r + x^{n-k} y^r z^k\Big).$$

where e is the base of the natural logarithm,

$$\binom{n}{r} = \frac{n!}{(n-r)!\,r!},$$

and $0! = 1$ and, for $n > 0$, $n! = n\,(n-1)!$.

Remark. The indices are restricted so that every term is defined: $n \geq 1$ ensures $(n+r-1)!$ makes sense, and $0 \leq k \leq n$ ensures $\binom{n}{k}$ makes sense. The $n = 0$ terms would vanish in any case, since $\binom{r-1}{r} = 0$ for $r \geq 1$.

Proof

Step 1: sum over k. Fix $n \geq 1$ and $r \geq 0$. By the binomial theorem,

$$\sum_{k=0}^{n}\binom{n}{k}y^k z^{n-k} = (y+z)^n, \quad \sum_{k=0}^{n}\binom{n}{k}x^k y^{n-k} = (x+y)^n, \quad \sum_{k=0}^{n}\binom{n}{k}x^{n-k} z^k = (x+z)^n.$$

Hence the sum over k equals

$$\frac{1}{(n+r-1)!}\binom{n+r-1}{r}\Big[x^r(y+z)^n + z^r(x+y)^n + y^r(x+z)^n\Big].$$

Step 2: group by total degree. Put $m = n+r \geq 1$. Then $n = m-r$, and $n \geq 1$ means $0 \leq r \leq m-1$. The series becomes

$$\sum_{m=1}^{\infty}\frac{1}{(m-1)!}\sum_{r=0}^{m-1}\binom{m-1}{r}\Big[x^r(y+z)^{m-r} + z^r(x+y)^{m-r} + y^r(x+z)^{m-r}\Big].$$

Step 3: sum over r. Write $(y+z)^{m-r} = (y+z)(y+z)^{m-1-r}$. By the binomial theorem,

$$\sum_{r=0}^{m-1}\binom{m-1}{r}x^r(y+z)^{m-1-r} = (x+y+z)^{m-1} = s^{m-1}.$$

Thus the three bracketed terms contribute $(y+z)s^{m-1}$, $(x+y)s^{m-1}$ and $(x+z)s^{m-1}$, which add to

$$\big[(y+z)+(x+y)+(x+z)\big]s^{m-1} = 2s\cdot s^{m-1} = 2s^m.$$

Step 4: sum over m.

$$\sum_{m=1}^{\infty} \frac{2s^m}{(m-1)!} = 2s \sum_{j=0}^{\infty} \frac{s^j}{j!} = 2s\, e^s. \qquad \blacksquare$$

Justification of the rearrangement. Replace x, y, z by $|x|, |y|, |z|$. Every term of the series is then nonnegative, and the computation above shows that the sum equals $2(|x| + |y| + |z|)\, e^{|x|+|y|+|z|} < \infty$. By Tonelli's theorem the order of summation is immaterial for nonnegative terms. Since the absolute value of each term at (x, y, z) is at most the corresponding term at $(|x|, |y|, |z|)$, the original series converges absolutely for all real x, y, z, and the regrouping in Steps 1–3 is valid.

Remark. The name is visible in Step 3: the three monomial families each contribute a sum of two of the variables times s^{m-1}, and the three pairs together contain each variable exactly twice, producing the factor $2s$.

The End